Luis Enrique Dalmau Coopat

Full-Stack AI Engineer · Computer Vision · YOLOv9

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SUMMARY

Full-Stack AI Engineer with **3+ years** building production systems across computer vision, ML pipelines, and full-stack web applications. Author of **sorba_ml_vision_restlib**, a proprietary computer-vision library used to train and deploy **YOLOv9** object-detection models for industrial IIoT predictive maintenance. B.S. in Computer Science, hands-on MLOps, and a teaching background in AI, ML, and distributed systems. Bilingual Spanish/English.

SKILLS

AI & Computer Vision: YOLOv9 · PyTorch · OpenCV · Object Detection · Image Segmentation · Auto-labeling (GroundingDINO, SAM, Florence-2) · Synthetic Data (Stable Diffusion) · Albumentations · TensorBoard · scikit-learn · pandas · NumPy
Backend: FastAPI · SQLAlchemy 2.0 · Alembic · Django · Pydantic · REST APIs · JWT/OAuth2 · APScheduler
Frontend: React 18 · Next.js 14 · TypeScript · Tailwind CSS · TanStack Query · Formik · Chart.js
Databases: PostgreSQL · MongoDB · MySQL · Redis
DevOps & Cloud: Docker · Docker Compose · Heroku · AWS S3 · Nginx · Git · GitLab CI · Linux
Testing & Quality: pytest · pytest-asyncio · pytest-cov · moto · httpx · ESLint · Prettier

EXPERIENCE

AI Engineer — Computer Vision
01/2024 – Present

SORBA.ai (Sorbotics LLC) · Jacksonville, USA (Remote)

- Authored **sorba_ml_vision_restlib** (300+ commits, v0.3.x): YOLOv9 training/inference, dataset management (YOLO/COCO/VOC), auto-labeling, synthetic data, GPU/VRAM scheduling.
- Built SGE vision nodes for real-time video: object detection, OCR (EasyOCR), pose/depth estimation, ByteTrack tracking.
- Resolved cross-environment CUDA/dependency issues and managed TLS for remote nodes.

Full-Stack Developer (Backend focus)
09/2024 – Present

Aerial Underground Construction Inc. · Kissimmee, FL, USA (Remote)

- Built logistics platform backend (FastAPI + SQLAlchemy + Alembic + PostgreSQL, 430+ commits): JWT auth, RBAC (3 roles), 20+ REST endpoints.
- Implemented commissions, daily/weekly/monthly reports, PDF/Excel invoices (ReportLab/openpyxl), AWS S3 (boto3), scheduled jobs (APScheduler).
- Contributed ~35% of a Next.js 14 + React + TypeScript frontend (310+ commits) with automated tests and Docker/Heroku deployment.

Teaching Assistant
01/2023 – 09/2024

University of Havana · Faculty of Mathematics & Computer Science

- Assisted Computer Networks, Machine Learning, AI, Operating Systems, and Distributed Systems; led practicals, graded exams, mentored students.

SELECTED PROJECTS

sorba_ml_vision_restlib — Proprietary CV library (author): YOLOv9 training, auto-labeling, synthetic data, GPU management.

Logistics Management System — FastAPI + Next.js + PostgreSQL; RBAC, reports, invoices, AWS S3.

Door Defect Detection (YOLOv9) — 539 annotated images, 4 classes, Roboflow augmentation.

LicitAI — FastAPI + React + MongoDB + Google Gemini (LLM) + Stripe.

Protectora RRSS — Pet-adoption diffusion: FastAPI + React + Telegram bot + SMTP.

Olympic Swimming Predictions — Data science, published on postdata.club.

EDUCATION & CERTIFICATIONS

B.S. in Computer Science — University of Havana (2017–2022). Thesis: "Graphic generation of signs of the Cuban Sign Language".

CS50's Introduction to AI with Python — HarvardX (edX) · **Data Science & Machine Learning** — Sololearn.

LANGUAGES

Spanish: Native · **English:** B2 (Professional Working Proficiency)